



Lesson 1: Hello, Computer Brains! What Coding Really Is



Big Idea

Coding is just giving clear, step-by-step instructions to a computer — like a recipe you write for a very literal robot friend.

Learning Objectives

By the end of this lesson, students will be able to:

- Explain, in their own words, what "code" and "coding" mean.
- Give an example of everyday step-by-step instructions (a recipe, a dance, brushing teeth).
- Understand that computers only do exactly what they are told — no more, no less.
- Name at least 2 things in daily life that run on code (phones, games, traffic lights, microwaves).
- Feel excited and confident that they can be coders too.

Teacher Preparation

- No computers needed yet for the first half — this lesson is mostly unplugged (away from the screen).
- Print or gather: one small everyday object (a sandwich, a paper airplane, or a picture of one).
- Have one tablet/computer per pair ready for the last 10 minutes if available, with scratch.mit.edu open (no login needed to look around).
- Review the "Human Robot" warm-up game below so you can play the "robot" convincingly and literally.

Required Files

- [Activities.md](#) (Human Robot game, Recipe-for-a-Robot worksheet activity)
- [Challenges.md](#)
- [Student_Lesson_Guide.md](#)

Teaching Flow (approx. 45 minutes)

1. Warm-up / Hook — 5 min
2. Concept Introduction — 10 min

3. Teacher Demonstration — 5 min
4. Guided Activity ("Human Robot") — 10 min
5. Mini Challenge — 5 min
6. Main Activity (Recipe for a Robot) — 8 min
7. Check for Understanding + Recap — 2 min

Warm-up / Hook (5 min)

 **Coding Mission: Wake Up the Class Robot!** Say to the class: "Good morning, coders! Today you all get a new superpower — the power to talk to computers in their OWN language. But first, let's find out what computers are REALLY like." Ask: "Have you ever asked a grown-up for a sandwich, and they made it wrong because you forgot to say 'no crusts'?" Let a couple of kids share funny examples. Say: "Computers are exactly like that. They do precisely what you tell them — nothing more, nothing sneaky, nothing they 'just guessed.' That's what coding is: giving super clear instructions."

Concept Introduction (10 min)

Introduce ideas gradually, with simple analogies: - **"What is code?"** — Code is a list of instructions written in a special language a computer can understand, kind of like a recipe card for a robot chef. - **"Why does a computer need EXACT instructions?"** — A computer has no imagination. If you tell it "make a sandwich" it has no idea what that means unless you say exactly: "pick up bread, put down bread, open peanut butter, spread peanut butter..." step by step. - **"Where is code hiding in real life?"** — Point out: the game they played on a tablet, the traffic light changing color, a microwave beeping after 30 seconds, a video game character jumping. All of these follow code someone wrote. -  **Robot Tip:** "Robots and computers are great helpers, but they need us humans to think of the plan first. That's YOUR job as a coder — you're the idea person, the computer is the helper who follows your plan perfectly."

Teacher Demonstration (5 min)

1. Hold up (or show a picture of) a peanut butter sandwich.
2. Say: "I am now a robot. I will only do exactly what you tell me, one step at a time."
3. Ask a student to give you ONE instruction at a time (e.g., "Pick up the bread").
4. Deliberately do something silly if the instruction is vague — for example, if they say "put the peanut butter on," pick up the whole jar and set it on top of the bread instead of spreading it.
5. Let the class laugh and correct their instruction to be more exact ("open the jar, dip the knife in, spread it on the bread").
6. Wrap up: "See? I did EXACTLY what you said — being a coder means being super clear."

Guided Coding — "Human Robot" Game (10 min)

Pair students up. One is the "coder," one is the "robot." The coder gives step-by-step spoken instructions (walk forward 3 steps, turn left, wave hand) to move the robot to a target spot in the room (a taped X or a chair). The robot only does exactly what is said, nothing implied. Swap roles halfway through.

 **Game Challenge:** Can your robot reach the target using only 5 instructions or fewer?

Mini Challenge (5 min)

In pairs, students write (or say aloud, if not yet writing independently) 4 instructions for "how to draw a happy face" — as if talking to a robot with no eyes for imagination, just following words exactly.

Main Activity / Project (8 min)

Recipe for a Robot worksheet: Students draw or write a 4–6 step "recipe" for a simple everyday task (brushing teeth, tying a shoe, feeding a pet). Remind them: no skipped steps, no "and then it just happens" — everything spelled out.

Fun Challenge (optional, harder) ★

Bonus Challenge: Write instructions for your robot to do something silly and fun, like "how to do a superhero pose" or "how to do a victory dance," using at least 6 exact steps.

Check for Understanding (spoken questions)

- "In your own words, what is coding?"
- "Why can't a computer just 'figure out' what you mean?"
- "Name one thing near you right now that might run on code."
- "What happens if we skip a step in our instructions?"

Common Student Mistakes (with fixes)

- **Mistake:** Giving vague instructions like "make it nice." **Fix:** Ask "nice HOW? What exactly should it look like/do?" Guide them to add specific details.
- **Mistake:** Skipping a step because "it's obvious." **Fix:** Remind them a computer has no common sense — walk through what happens if that step is missing.

- **Mistake:** Getting frustrated when the "robot" (teacher/partner) does something silly. **Fix:** Reframe it as funny and helpful — "That silly mistake taught us exactly what to fix!"

Differentiation

- **Struggling support:** Let them draw pictures in order instead of writing words; give them a 3-step task instead of 4–6.
- **Standard:** Follow the plan as written above.
- **Extension for advanced learners:** Challenge them to write instructions for a two-part task (e.g., "make a sandwich AND pour a glass of juice") and to spot which step, if skipped, would cause the funniest mistake.

Recap

"Today you learned that coding is just giving super clear, step-by-step instructions — and that computers need EVERY step, because they can't guess. You've already started thinking like a coder!"

Take-Home / Creative Challenge

Ask a grown-up at home to be your "robot" for one task tonight (like setting the table or getting dressed for bed). Give them only exact spoken instructions, one at a time, and see what happens if you forget a step!

Teacher Talking Points

- "I love how carefully you're all thinking about this — that's exactly what real coders do every single day."
- "It's okay if your robot did something silly — that's not a mistake, that's a clue that helps us fix our instructions!"
- "You don't need a computer to be a coder yet — you just need to think in clear steps, and you're already doing it."
- "Coding is really just clear thinking, written down so a computer can follow along."



Victory Moment

At the end of class, have every student say one thing "code" controls in their own house. Celebrate: "Look at that — you're already spotting code everywhere. You're officially coder detectives!"